

MATHEMATICS

TARGET : JEE- Advanced

CAPS-5

SEQUENCE & SERIES

SCQ (Single Correct Type) :

- The sequence $a_1, a_2, a_3, \dots$ satisfies $a_1 = 1, a_2 = 2$ and $a_{n+2} = \frac{2}{a_{n+1}} + a_n, n = 1, 2, 3, \dots$ the value of $\frac{2^{2009}}{2011} \cdot a_{2012}$ is
 (A) $^{2010}C_{1005}$ (B) $^{2011}C_{1006}$ (C) $^{2011}C_{1005}$ (D) $^{2012}C_{1006}$
- Let $a_1, a_2, a_3, \dots, a_n$ be in G.P. If the area bounded by the curves $y^2 = 4a_n x$ and $y^2 = 4a_n(a_n - x)$ be A_n , then the sequence $A_1, A_2, A_3, \dots, A_n$ are in
 (A) A. P. (B) G. P. (C) H. P. (D) None of these
- The sum of first 'n' terms of the series $\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \dots$ is
 (A) 2^{n-1} (B) $1 - 2^{-n}$ (C) $2^{-n} - n + 1$ (D) $2^{-n} + n - 1$
- Find the value of $\sum_{r=1}^n \sum_{s=1}^n \delta_{rs} 2^r 3^s$ where $\begin{cases} \delta_{rs} = 0, & \text{if } r \neq s \\ \delta_{rs} = 1, & \text{if } r = s \end{cases}$
 (A) $\frac{6}{5}(6^n - 1)$ (B) $6^n - 1$ (C) $\frac{1}{5}(6^n - 1)$ (D) None of these
- Value of $\sum_{r=1}^{\infty} \frac{1}{r(r+1)(r+2)(r+3)} =$ —
 (A) 1 (B) 1/2 (C) 1/18 (D) 0
- If α, γ are the roots of $t_1 x^2 - 4x + 1 = 0$ and β, δ are the roots of $t_2 x^2 - 6x + 1 = 0$ and $\alpha, \beta, \gamma, \delta$ are in H.P. then
 (A) $-t_1 + t_2 = 5$ (B) $t_1 + t_2 = 12$ (C) $t_1 = 8$ (D) $t_2 = 5$
- If $1^2 + 2^2 + 3^2 + \dots + 2003^2 = (2003)(4007)(334)$ and
 $(1)(2003) + (2)(2002) + (3)(2001) + \dots + (2003)(1) = (2003)(334)(x)$, then x equals
 (A) 2005 (B) 2004 (C) 2003 (D) 2001

MCQ (One or more than one correct) :

- Consider the sequence a_n given by $a_1 = \frac{1}{2}, a_{n+1} = a_n^2 + a_n$,
 Let $S_n = \frac{1}{a_1 + 1} + \frac{1}{a_2 + 1} + \dots + \frac{1}{a_n + 1}$ then find the value of $[S_{2012}]$, where $[.]$ denotes greatest integer function.
 (A) 1 (B) $[e / 2]$ (C) $[e]$ (D) $[\pi - 1]$



9. Given that $x + y + z = 15$ when a, x, y, z, b are in A. P. and $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{5}{3}$ when a, x, y, z, b are in H. P. Then
- (A) G. M. of a and b is 3 (B) one possible value of $(a + 2b)$ is 11
- (C) A. M. of a and b is 5 (D) H. M. of a and b is $\frac{9}{5}$
10. The product of two positive real numbers a and b is 192. The quotient of A.M. by H.M. of their G.C.D and L.C.M is $\frac{169}{48}$. The smaller of a and b can be
- (A) 2 (B) 4 (C) 6 (D) 12
11. Let a, b, c are distinct real numbers such that expression $ax^2 + bx + c, bx^2 + cx + a$ and $cx^2 + ax + b$ are always positive then possible value(s) of $\frac{a^2 + b^2 + c^2}{ab + bc + ca}$ may be:
- (A) 1 (B) 2 (C) 3 (D) 4
12. For $\triangle ABC$, if $81 + 144a^4 + 16b^4 + 9c^4 = 144abc$, (where notations have their usual meaning), then
- (A) $a > b > c$ (B) $A < B < C$
- (C) Area of $\triangle ABC = \frac{3\sqrt{3}}{8}$ (D) Triangle ABC is right angled
13. Let $x, y, z \in \left(0, \frac{\pi}{2}\right)$ are first three consecutive terms of an arithmetic progression such that $\cos x + \cos y + \cos z = 1$ and $\sin x + \sin y + \sin z = \frac{1}{\sqrt{2}}$, then which of the following is/are correct?
- (A) $\cot y = \sqrt{2}$ (B) $\cos(x - y) = \frac{\sqrt{3} - \sqrt{2}}{2\sqrt{2}}$
- (C) $\tan 2y = \frac{2\sqrt{2}}{3}$ (D) $\sin(x - y) + \sin(y - z) = 0$
14. $a_1, a_2, \dots$ are distinct terms of an A.P. We call (p, q, r) an increasing triad if a_p, a_q, a_r are in G.P. where $p, q, r \in \mathbb{N}$ such that $p < q < r$. If $(5, 9, 16)$ is an increasing triad, then which of the following option is/are correct
- (A) If a_1 is a multiple of 4 then every term of the A.P. is an integer
- (B) $(85, 149, 261)$ is an increasing triad
- (C) If the common difference of the A.P. is $\frac{1}{4}$, then its first term is $\frac{1}{3}$
- (D) Ratio of the $(4k + 1)^{\text{th}}$ and $4k^{\text{th}}$ term can be 4

Numerical based Questions :

15. If $\frac{25}{k} = 1^2 - \frac{2^2}{5} + \frac{3^2}{5^2} - \frac{4^2}{5^3} + \frac{5^2}{5^4} - \frac{6^2}{5^5} + \dots \infty$, then find the value of k
16. A man arranges to pay off a debt of Rs. 3600 by 40 annual installments which form an arithmetic series. When 30 of the installments are paid he dies leaving a third of the debt unpaid. Find the value of the first installment.
17. If $x > 0$, and $\log_2 x + \log_2(\sqrt{x}) + \log_2(\sqrt[4]{x}) + \log_2(\sqrt[8]{x}) + \log_2(\sqrt[16]{x}) + \dots = 4$, then find x.
18. If $x_i > 0$, $i = 1, 2, \dots, 50$ and $x_1 + x_2 + \dots + x_{50} = 50$, then find the minimum value of $\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}}$.
19. The number of terms in an A.P. is even ; the sum of the odd terms is 24, sum of the even terms is 30, and the last term exceeds the first by $10\frac{1}{2}$; find the number of terms.
20. If a, b, c are in GP, $a - b$, $c - a$, $b - c$ are in HP, then the value of $a + 4b + c$ is
21. If $S = \frac{5}{13} + \frac{55}{(13)^2} + \frac{555}{(13)^3} + \frac{5555}{(13)^4} + \dots$ up to ∞ , then find the value of 36S.
22. If $S = \frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \dots \infty$, then find the value of 14S.

Matrix Match Type :

23. Match Column I with Column II:

Column-I		Column-II	
A	$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}}$ is equal to	p	1
B	$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{6^k}{(3^k - 2^k)(3^{k+1} - 2^{k+1})}$ is equal to	q	2
C	$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^2 - \frac{1}{2}}{k^4 + \frac{1}{4}}$ is equal to	r	3
D	$x_1 = \frac{1}{2}$ and $x_{k+1} = x_k^2 + x_{k-1}$ then $T = \sum_{k=1}^n \frac{1}{x_k + 1}$ then [T] is equal to (where [.] denotes G.I.F.)	s	-1
		t	0

Code :

(A) A-p; B-r; C-q; D-s

(B) A-r; B-p; C-s; D-r

(C) A-p; B-q; C-p; D-p

(D) A-r; B-p; C-q; D-r

Subjective Type Questions :

24. In an A.P. of which 'a' is the 1st term, if the sum of the 1st 'p' terms is equal to zero, show that the sum of the next 'q' terms is $-\frac{a(p+q)q}{p-1}$.
25. The sum of first p-terms of an A.P. is q and the sum of first q terms is p, find the sum of first (p + q) terms.

ANSWER KEY (CAPS-5)

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|---------|----------|----------------------------|--------------|-----------|-----------|-----------|
| 1. (A) | 2. (B) | 3. (D) | 4. (A) | 5. (C) | 6. (A) | 7. (A) |
| 8. (AB) | 9. (ABC) | 10. (BD) | 11. (BC) | 12. (BCD) | 13. (ABC) | 14. (ABC) |
| 15. 54 | 16. 51 | 17. 4 | 18. 50 | 19. 8 | 20. 0 | 21. 65 |
| 22. 7 | 23. (C) | 24. $-\frac{a(p+q)q}{p-1}$ | 25. $-(p+q)$ | | | |